Team number: 7 **Assigned Phase 2 sub converted:** Team 16 (Wife Carrying Championships)

Project Phase 3

Relational Model and Normalization

1 Stage A: ER/EER-to-Relational Mapping

We mapped Team 16's final Phase 2 EER diagram, including every fix listed in their Section 2.4. Team 16 did not draw foreign-key attributes on their diagram (their fix K1). We added each one back as the relationship it came from (R1, R3, R4, R7, R13, R14).

Strong entity types

COUNTRY, PARTICIPANT, CATEGORY, COMPETITION_EDITION, COMPETITION_ENTRY, OFFICIAL and RESULT_CORRECTION_LOG each became a relation with all their simple attributes. Composite attributes were replaced by their simple components:

- PARTICIPANT.name → first_name, last_name
- OFFICIAL.name → first_name, last_name
- COMPETITION_EDITION.date_range → start_date, end_date
- CATEGORY.rules → eligibility_rules, race_rules

Primary keys are country_code, participant_id, category_id, edition_year, entry_id, official_id and correction_id. Alternate keys are kept as UNIQUE constraints:

- COUNTRY.country_name
- CATEGORY.category_name
- COMPETITION_ENTRY (edition_year, category_id, bib_number)

Derived attributes are not stored and will be computed with queries or views:

- age_at_competition (on R12 PARTICIPATES_IN_EDITION, per participant–edition) and editions_entered (PARTICIPANT)
- total_participants and total_entries (COMPETITION_EDITION)
- entrant_count (HEAT)
- final_time and time_diff (RECORDS_RESULT)

Weak entity types

RACE (owner COMPETITION_EDITION, identifying relationship R6) became RACE with edition_year as a foreign key to COMPETITION_EDITION. Its primary key is (edition_year, race_no).

HEAT (owner RACE, identifying relationship R8) became HEAT with (edition_year, race_no) as a composite foreign key to RACE. Its primary key is (edition_year, race_no, heat_no). Because HEAT's owner is itself weak, HEAT inherits the full two-part key of RACE.

Binary relationships (1:1, 1:N, M:N)

- **1:1:** The diagram has no 1:1 relationships.

- **1:N:** These were mapped with a foreign key on the N side:
  - R1 REPRESENTS → PARTICIPANT.country_code
  - R3 HAS_ENTRY → COMPETITION_ENTRY.edition_year
  - R4 ENTERED_IN → COMPETITION_ENTRY.category_id
  - R7 OF_CATEGORY → RACE.category_id
  - R14 MADE_BY → RESULT_CORRECTION_LOG.official_id
  - R6 and R8 are covered by the weak-entity mapping above.
- **R11 WINS:** Team 16 changed this to 1:N (COMPETITION_ENTRY (0,1), COMPETITION_EDITION (0,N)). We mapped it as a separate relation WINS(entry_id, edition_year) with primary key entry_id. Most entries never win, so a foreign-key column in COMPETITION_ENTRY would be mostly NULL.
- **M:N:** Each became its own relation whose primary key combines both participants' keys:
  - R2 HAS_CATEGORY(edition_year, category_id)
  - R5 PARTICIPATES_IN(entry_id, participant_id, participant_role). The relationship attribute participant_role is stored here.
  - R9 SUPERVISES(official_id, edition_year, race_no, heat_no)
  - R12 PARTICIPATES_IN_EDITION(participant_id, edition_year)
  - R15 ASSIGNED_TO(entry_id, edition_year, race_no, heat_no). This is the relationship Team 16 added in Phase 2 (fix M1).

Multi-valued attributes

Attribute	Option	Max values	Reason
PARTICIPANT.contact_numbers	(b) Separate relation PARTICIPANT_CONTACT(participant_id, contact_number)	—	The requirements give no fixed maximum, so the number of contact numbers per participant can vary.
CATEGORY.aliases	(b) Separate relation CATEGORY_ALIAS(category_id, alias)	—	A category can have any number of alternative names, with no known upper limit.

Relationships of degree $n > 2$

R10 RECORDS_RESULT is ternary (COMPETITION_ENTRY, HEAT, OFFICIAL). It became the relation RECORDS_RESULT with:

- foreign keys entry_id (→ COMPETITION_ENTRY), (edition_year, race_no, heat_no) (→ HEAT) and official_id (→ OFFICIAL);
- the relationship attributes raw_time, penalty_seconds, rank, result_status and recording_timestamp.

The primary key is (entry_id, edition_year, race_no, heat_no), not all three participants' keys. Team 16's fix M9 states that an (entry, heat) pair has at most one result, so official_id is an ordinary foreign key.

Specialization / generalization

COMPETITION_ENTRY is specialized into PAIR_ENTRY and TEAM_ENTRY (disjoint, total). We used **option 8A**, which creates one relation for the superclass and one for each subclass:

- COMPETITION_ENTRY holds all shared attributes.
- PAIR_ENTRY(entry_id, carrier_id, carried_id) and TEAM_ENTRY(entry_id, team_name) each use entry_id as both primary key and foreign key to COMPETITION_ENTRY.
- carrier_id and carried_id are foreign keys to PARTICIPANT, kept as in the assigned diagram.

We chose 8A over 8C/8D (a single relation with a type attribute) because the subclasses have different attributes, so one relation would leave many NULLs. We chose it over 8B (subclass relations only) because COMPETITION_ENTRY is the target of many relationships (R5, R10, R11, R15) and needs its own relation for those foreign keys. 8A does not enforce the disjoint, total constraint by itself, so it will be enforced with a check or trigger.

Recursive relationships

The assigned diagram has no recursive relationships.

Union types (categories) and other constructs

The assigned diagram has no union types.

It does use **aggregation**. Team 16 drew R10 as an aggregate so that R13 CORRECTS could relate RESULT_CORRECTION_LOG to a recorded result (fix M7). Following the textbook, we treated the aggregate as the relation RECORDS_RESULT and mapped R13, which is 1:N with the log on the N side, as a foreign key in RESULT_CORRECTION_LOG. That foreign key is (entry_id, edition_year, race_no, heat_no) and references RECORDS_RESULT's primary key.

Notation

Underlined attributes form the primary key, and each foreign key has an arrow to the primary key it references.

Deviations: arrows are coloured by the referenced relation, and a bracket under several attributes marks a composite foreign key.

Stage A snapshot

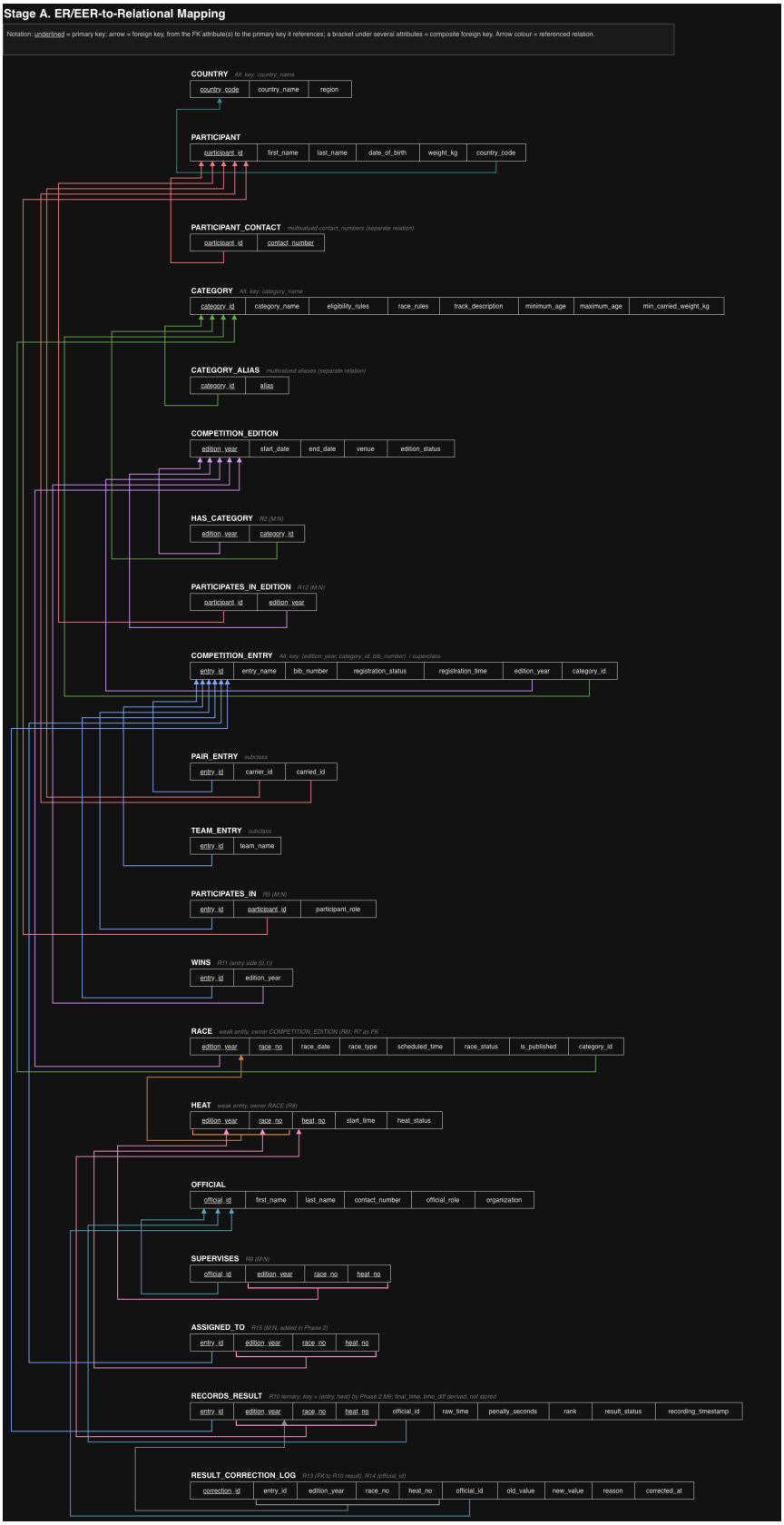


Diagram link: `https://drive.google.com/file/d/1lZHS0WmBY_4_edJjbHpK4qps9e1XLwZm/view`

2 Stage B: First Normal Form (1NF)

No conversion was needed; no snapshot is submitted for this stage.

We checked every Stage A relation for composite values, lists of values in one column, and repeating groups. All attributes are atomic: composite attributes (name, date_range, rules) were already split into simple columns in Stage A, and the two multi-valued attributes (contact_numbers, aliases) were already mapped with option (b) to PARTICIPANT_CONTACT and CATEGORY_ALIAS. No relation has a repeating group, so the Stage A model is already in 1NF and Stage C starts from it unchanged.

3 Stage C: Second Normal Form (2NF)

Stage C starts from the Stage B model (identical to Stage A). A relation is in 2NF if it is in 1NF and every non-prime attribute depends on the whole of *every* candidate key. We used this general definition, which considers all candidate keys, not only the primary key.

Only two relations needed a change: RECORDS_RESULT and ASSIGNED_TO.

- **The dependency.** In Stage A both relations were given the key (entry_id, edition_year, race_no, heat_no). But each COMPETITION_ENTRY belongs to exactly one edition (R3, total (1,1)), so entry_id → edition_year.
- **The key.** The four-attribute key is therefore not minimal. The candidate key is (entry_id, race_no, heat_no), and edition_year is non-prime.
- **The violation.** edition_year depends on only part of the candidate key (entry_id), which is a partial dependency.
- **The fix.** The decomposition would give R1(entry_id, edition_year) and R2(the remaining attributes). R1 is already contained in COMPETITION_ENTRY, so edition_year is removed from both relations and recovered through entry_id.
  - RECORDS_RESULT(<u>entry_id</u>, <u>race_no</u>, <u>heat_no</u>, official_id, raw_time, penalty_seconds, rank, result_status, recording_timestamp)
  - ASSIGNED_TO(<u>entry_id</u>, <u>race_no</u>, <u>heat_no</u>)
- **Foreign keys.** The heat is still identified by (entry_id, race_no, heat_no), with edition_year taken from entry_id. A composite FK (edition_year, race_no, heat_no) → HEAT can no longer be declared directly, so heat existence is enforced by a trigger or check.
- **RESULT_CORRECTION_LOG.** Its R13 foreign key is now (entry_id, race_no, heat_no) → RECORDS_RESULT, because RECORDS_RESULT's primary key changed. Its only candidate key is the single attribute correction_id, so it has no partial dependency. Its edition_year column depends on entry_id, which is a transitive dependency and is handled in Stage D.

Relations needing no change:

- COUNTRY, PARTICIPANT, CATEGORY, COMPETITION_EDITION, OFFICIAL, PAIR_ENTRY, TEAM_ENTRY and WINS have only single-attribute candidate keys, so a partial dependency is impossible.
- COMPETITION_ENTRY also has the composite candidate key (edition_year, category_id, bib_number). Its non-prime attributes (entry_name, registration_status, registration_time) depend on the whole entry, not on any proper subset of that key, so there is no partial dependency.

- PARTICIPANT_CONTACT, CATEGORY_ALIAS, HAS_CATEGORY, PARTICIPATES_IN_EDITION and SUPERVISES are all-key relations with no non-prime attributes.
- PARTICIPATES_IN has the key (entry_id, participant_id), and participant_role depends on both, since a role is specific to a participant within an entry.
- RACE has the key (edition_year, race_no) and HEAT has the key (edition_year, race_no, heat_no). race_no and heat_no are unique only within their owner, so no attribute depends on a proper subset of the key.

Relations after Stage C

Relation	Primary key	Other attributes and foreign keys
COUNTRY	country_code	country_name (alt. key), region
PARTICIPANT	participant_id	first_name, last_name, date_of_birth, weight_kg, country_code (FK)
PARTICIPANT_CONTACT	participant_id, contact_number	participant_id (FK)
CATEGORY	category_id	category_name (alt. key), eligibility_rules, race_rules, track_description, minimum_age, maximum_age, min_carried_weight_kg
CATEGORY_ALIAS	category_id, alias	category_id (FK)
COMPETITION_EDITION	edition_year	start_date, end_date, venue, edition_status
HAS_CATEGORY	edition_year, category_id	both are FKs
PARTICIPATES_IN_EDITION	participant_id, edition_year	both are FKs
COMPETITION_ENTRY	entry_id	entry_name, bib_number, registration_status, registration_time, edition_year (FK), category_id (FK); alt. key (edition_year, category_id, bib_number)
PAIR_ENTRY	entry_id	entry_id (FK), carrier_id (FK), carried_id (FK)
TEAM_ENTRY	entry_id	entry_id (FK), team_name
PARTICIPATES_IN	entry_id, participant_id	participant_role; both key attributes are FKs
WINS	entry_id	entry_id (FK), edition_year (FK)
RACE	edition_year, race_no	race_date, race_type, scheduled_time, race_status, is_published, category_id (FK)
HEAT	edition_year, race_no, heat_no	start_time, heat_status; (edition_year, race_no) FK → RACE
OFFICIAL	official_id	first_name, last_name, contact_number, official_role, organization
SUPERVISES	official_id, edition_year, race_no, heat_no	official_id (FK), (edition_year, race_no, heat_no) FK → HEAT
ASSIGNED_TO (changed)	entry_id, race_no, heat_no	entry_id (FK); heat existence enforced by trigger
RECORDS_RESULT (changed)	entry_id, race_no, heat_no	entry_id (FK), official_id (FK), raw_time, penalty_seconds, rank, result_status, recording_timestamp; heat existence enforced by trigger

Relation	Primary key	Other attributes and foreign keys
RESULT_CORRECTION_LOG	correction_id	(entry_id, race_no, heat_no) FK → RECORDS_RESULT, edition_year, official_id (FK), old_value, new_value, reason, corrected_at

Stage C snapshot

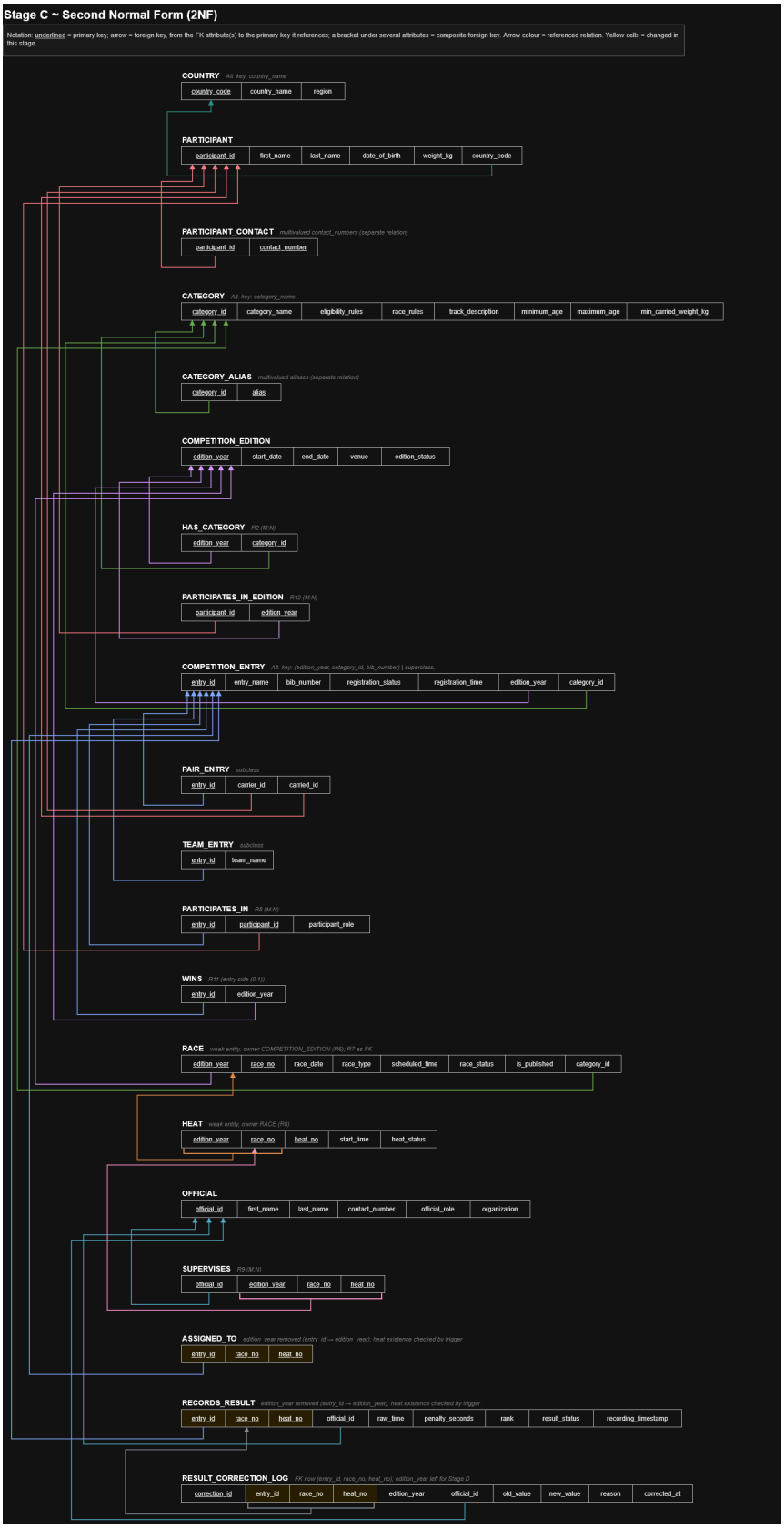


Diagram link: `https://drive.google.com/file/d/1VhCOreYHSX3CaGmQ18eDJo94bDJE58a3/view?usp=sharing`

4 Stage D: Third Normal Form (3NF)

Compared to Stage C, only RESULT_CORRECTION_LOG was changed, to remove a transitive dependency.

In Stage C, correction_id → entry_id and entry_id → edition_year. entry_id is not a superkey of RESULT_CORRECTION_LOG (one result can be corrected many times), and edition_year is non-prime, so edition_year depended transitively on the candidate key correction_id through entry_id. This violates 3NF. Since edition_year can be retrieved from COMPETITION_ENTRY using entry_id, storing it again was redundant.

Functional dependencies responsible for the change:

- correction_id → entry_id
- entry_id → edition_year (entry_id is not a superkey; edition_year is non-prime)
- therefore correction_id → edition_year is transitive.

Fix. The decomposition would give (entry_id, edition_year), which is already in COMPETITION_ENTRY, so edition_year was removed from RESULT_CORRECTION_LOG. All other attributes and relations remain unchanged.

Resulting relation:
RESULT_CORRECTION_LOG(correction_id, entry_id, race_no, heat_no, official_id, old_value, new_value, reason, corrected_at), with FK (entry_id, race_no, heat_no) → RECORDS_RESULT and FK official_id → OFFICIAL.

Reason for no further changes. All other relations already satisfy 3NF: for every non-trivial functional dependency X → A in them, either X is a superkey or A is a prime attribute (see Section 5). Hence no further decomposition was required.

Stage D snapshot

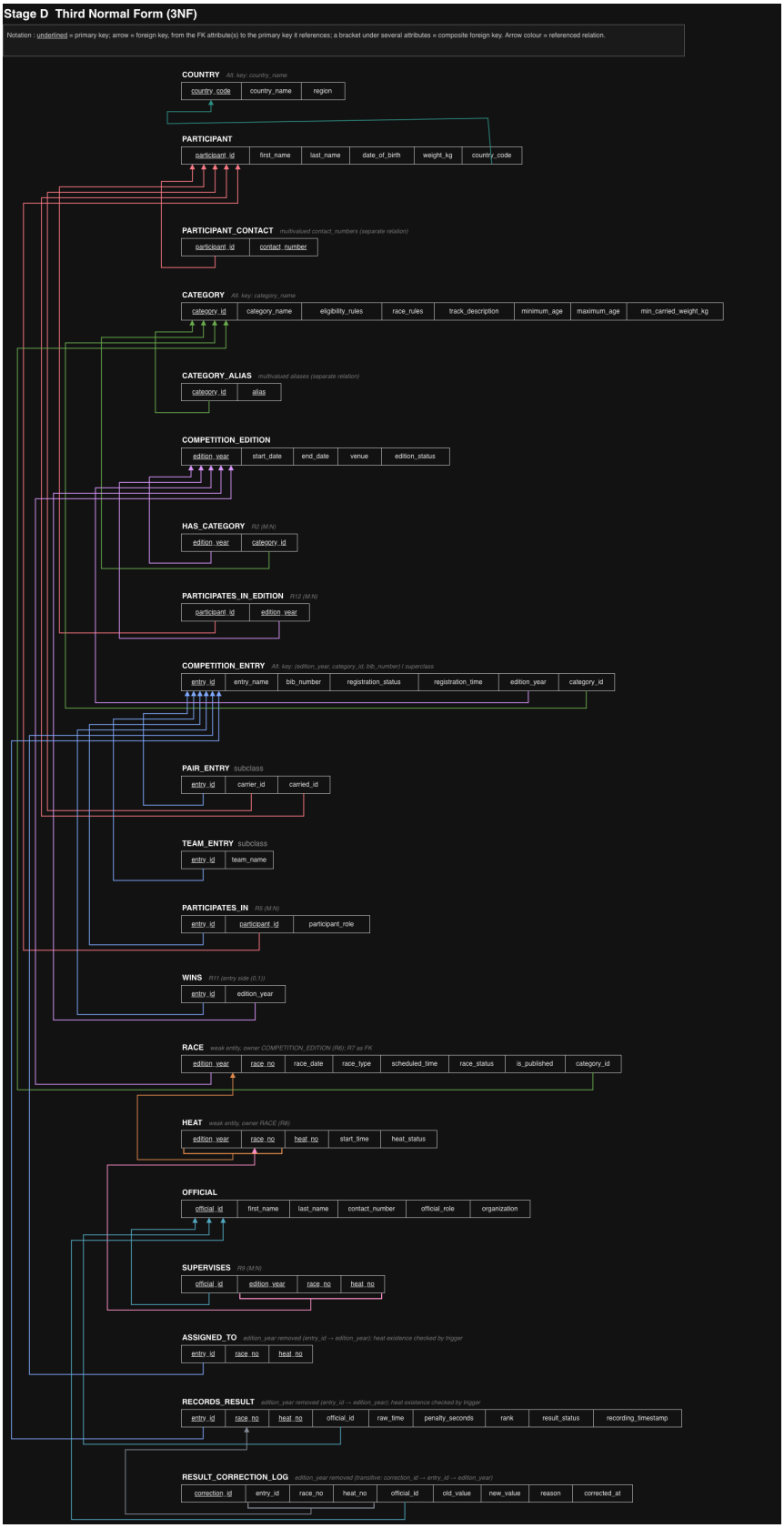


Diagram link: `https://drive.google.com/file/d/1lb-rSojVEFH2MLFae3VTZkC777xC5Ndt/view?usp=sharing`

5 Functional Dependencies

Candidate keys and the functional dependencies used or checked for every relation. All dependencies follow from the meaning of the Wife Carrying World Championships data.

Relations changed

RECORDS_RESULT (changed in Stage C)
Candidate key: (entry_id, race_no, heat_no)

- entry_id → edition_year (partial dependency; edition_year removed in Stage C)
- entry_id, race_no, heat_no → official_id, raw_time, penalty_seconds, rank, result_status, recording_timestamp

ASSIGNED_TO (changed in Stage C)
Candidate key: (entry_id, race_no, heat_no)

- entry_id → edition_year (partial dependency; edition_year removed in Stage C)

RESULT_CORRECTION_LOG (changed in Stage D)
Candidate key: correction_id

- correction_id → entry_id, race_no, heat_no, official_id, old_value, new_value, reason, corrected_at
- entry_id → edition_year (transitive dependency; edition_year removed in Stage D)

Relations checked, no change

- **COUNTRY** — candidate keys: country_code; country_name
  - country_code → country_name, region
  - country_name → country_code
- **PARTICIPANT** — candidate key: participant_id
  - participant_id → first_name, last_name, date_of_birth, weight_kg, country_code
- **CATEGORY** — candidate keys: category_id; category_name
  - category_id → category_name, eligibility_rules, race_rules, track_description, minimum_age, maximum_age, min_carried_weight_kg
  - category_name → category_id
- **COMPETITION_EDITION** — candidate key: edition_year
  - edition_year → start_date, end_date, venue, edition_status
- **COMPETITION_ENTRY** — candidate keys: entry_id; (edition_year, category_id, bib_number)
  - entry_id → entry_name, bib_number, registration_status, registration_time, edition_year, category_id
  - edition_year, category_id, bib_number → entry_id
- **PAIR_ENTRY** — candidate key: entry_id

  - entry_id → carrier_id, carried_id

- **TEAM_ENTRY** — candidate key: entry_id
  - entry_id → team_name
- **PARTICIPATES_IN** — candidate key: (entry_id, participant_id)
  - entry_id, participant_id → participant_role
- **WINS** — candidate key: entry_id
  - entry_id → edition_year
- **RACE** — candidate key: (edition_year, race_no)
  - edition_year, race_no → race_date, race_type, scheduled_time, race_status, is_published, category_id
- **HEAT** — candidate key: (edition_year, race_no, heat_no)
  - edition_year, race_no, heat_no → start_time, heat_status
- **OFFICIAL** — candidate key: official_id
  - official_id → first_name, last_name, contact_number, official_role, organization
- **All-key relations** (PARTICIPANT_CONTACT, CATEGORY_ALIAS, HAS_CATEGORY, PARTICIPATES_IN_EDITION, SUPERVISES): the candidate key is the whole attribute set, so only trivial dependencies hold.

6 Changes to the Assigned EER Diagram

We made no changes to Team 16's final EER diagram. All of Team 16's own Phase 2 fixes (Section 2.4 of their submission) were applied as they listed them, and the foreign-key attributes they left off the diagram (fix K1) were added back during mapping, as they intended.

7 GenAI Usage Statement

We used AI to check our mapping and normalization reasoning, and convert the document to LaTeX]. Every relation, key, functional dependency and decomposition was reviewed by us.